ENGR 311 — Problems

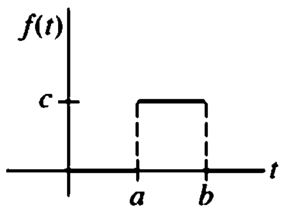
Section 4.1 · ed. 5 p.213

In Problems 1–18, use Definition 4.1.1 to find $\mathscr{L}\{f(t)\}$.

3. $f(t) = \begin{cases} t, & 0 \le t < 1 \\ 1, & t \ge 1 \end{cases}$

6. $f(t) = \begin{cases} \sin t, & 0 \le t < \pi/2 \\ 0, & t \ge \pi/2 \end{cases}$

10.



14. $f(t) = t^2 e^{-2t}$

Section 4.2 · ed. 5 p.221

In Problems 1–30, use Theorem 4.2.1 to find the given inverse

5. $\mathscr{L}^{-1}\left\{\dfrac{(s+1)^3}{s^4}\right\}$

6. $\mathscr{L}^{-1}\left\{\dfrac{(s+2)^2}{s^3}\right\}$

13. $\mathscr{L}^{-1}\left\{\dfrac{4s}{4s^2+1}\right\}$

16. $\mathscr{L}^{-1}\left\{\dfrac{s+1}{s^2+2}\right\}$

In Problems 31–40, use the Laplace transform to solve the given initial-value problem.

36. $y'' - 4y' = 6e^{3t} - 3e^{-t}, \quad y(0) = 1, \; y'(0) = -1$

40. $y'' + 9y = e^t, \quad y(0) = 0, \; y'(0) = 0$

Section 4.3 · ed. 5 p.229, p.230

In Problems 1–20, find either $F(s)$ or $f(t)$, as indicated.

5. $\mathscr{L}\{t(e^t + e^{2t})^2\}$

12. $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)^4}\right\}$

In Problems 21–30, use the Laplace transform

22. $y' - y = 1 + te^t, \quad y(0) = 0$

24. $y'' - 4y' + 4y = t^3e^{2t}, \quad y(0) = 0, \quad y'(0) = 0$

In Problems 37–48, find either $F(s)$ or $f(t)$, as indicated.

41. $\mathscr{L}\{\cos 2t\, \mathscr{U}(t - \pi)\}$

45. $\mathscr{L}^{-1}\left\{\dfrac{e^{-\pi s}}{s^2 + 1}\right\}$

Section 4.4 · ed. 5 p.240, p.241

In Problems 1–8, use Theorem 4.4.1 to evaluate the given Laplace transform.

4. $\mathscr{L}\{t \sinh 3t\}$

7. $\mathscr{L}\{te^{2t} \sin 6t\}$

8. $\mathscr{L}\{te^{-3t} \cos 3t\}$

In Problems 9–14, use the Laplace transform to solve the given

10. $y' - y = te^t \sin t, \quad y(0) = 0$

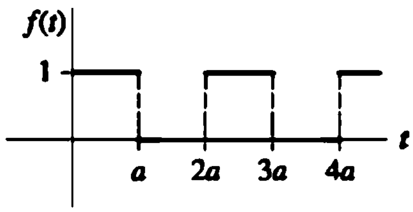
In Problems 37–46, use the Laplace transform to solve the given integral equation or integrodifferential equation.

45. $y'(t) = 1 - \sin t - \displaystyle\int_0^t y(\tau)\,d\tau, \quad y(0) = 0$

46. $\dfrac{dy}{dt} + 6y(t) + 9\displaystyle\int_0^t y(\tau)\,d\tau = 1, \quad y(0) = 0$

In Problems 51–56, use Theorem 4.4.3 to find the Laplace transform of the given periodic function.

52.



Square wave

Section 4.5 · ed. 5 p.245

In Problems 1–12, use the Laplace transform to solve the given differential equation subject to the indicated initial conditions.

3. $y'' + y = \delta(t - 2\pi), \quad y(0) = 0, y'(0) = 1$

9. $y'' + 4y' + 5y = \delta(t - 2\pi), \quad y(0) = 0, y'(0) = 0$

12. $y'' - 7y' + 6y = e^t + \delta(t - 2) + \delta(t - 4), \quad y(0) = 0, y'(0) = 0$

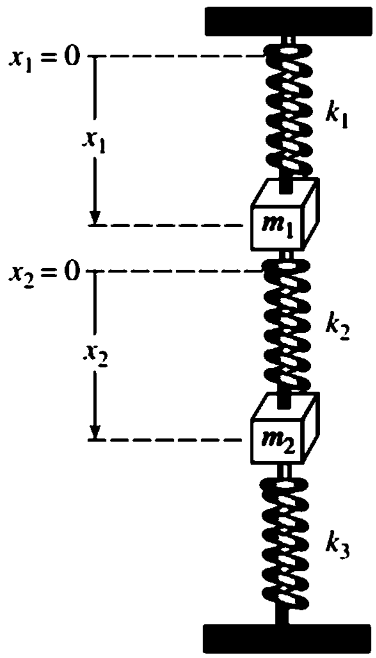
Section 4.6 · ed. 5 p.248, p.249

In Problems 1–12, use the Laplace transform to solve the given system of differential equations.

4.
$$\frac{dx}{dt} + 3x + \frac{dy}{dt} = 1$$
$$\frac{dx}{dt} - x + \frac{dy}{dt} - y = e^t$$
$$x(0) = 0, \; y(0) = 0$$

10.
$$\frac{dx}{dt} - 4x + \frac{d^3y}{dt^3} = 6 \sin t$$
$$\frac{dx}{dt} + 2x - 2\frac{d^3y}{dt^3} = 0$$
$$x(0) = 0, \; y(0) = 0,$$
$$y'(0) = 0, \; y''(0) = 0$$

14. Derive the system of differential equations describing the straight-line vertical motion of the coupled springs shown in equilibrium in **FIGURE 4.6.4**. Use the Laplace transform to solve the system when $k_1 = 1, k_2 = 1, k_3 = 1, m_1 = 1, m_2 = 1$ and $x_1(0) = 0, x_1'(0) = -1, x_2(0) = 0, x_2'(0) = 1$.



Section 12.1 · ed. 5 p.659

In Problems 1–6, show that the given functions are orthogonal on the indicated interval.

3. $f_1(x) = e^x, f_2(x) = xe^{-x} - e^{-x}$; $[0, 2]$

6. $f_1(x) = e^x, f_2(x) = \sin x$; $[\pi/4, 5\pi/4]$

In Problems 7–12, show that the given set of functions is orthogonal on the indicated interval. Find the norm of each function in the set.

10. $\left\{\sin \frac{n\pi}{p} x\right\}$, $n = 1, 2, 3, \ldots$; $[0, p]$

18. From Problem 1 we know that $f_1(x) = x$ and $f_2(x) = x^2$ are orthogonal on $[-2, 2]$. Find constants c_1 and c_2 such that $f_3(x) = x + c_1x^2 + c_2x^3$ is orthogonal to both f_1 and f_2 on the same interval.

21. A real-valued function is said to be **periodic** with period $T \neq 0$ if $f(x + T) = f(x)$ for all x in the domain of f. If T is the smallest positive value for which $f(x + T) = f(x)$ holds, then T is called the **fundamental period** of f. In Problems 21–26, determine the fundamental period T of the given function.

(a) $f(x) = \cos 2\pi x$

(b) $f(x) = \sin \dfrac{4}{L} x$

(c) $f(x) = \sin x + \sin 2x$

(d) $f(x) = \sin 2x + \cos 4x$

(e) $f(x) = \sin 3x + \cos 2x$

(f) $f(x) = A_0 + \sum_{n=1}^{\infty}\left(A_n \cos \dfrac{n\pi}{p} x + B_n \sin \dfrac{n\pi}{p} x\right)$, A_n and B_n depend only on n

Section 12.2 · ed. 5 p.664

In Problems 1–16, find the Fourier series of f on the given

10. $f(x) = \begin{cases} 0, & -\pi/2 < x < 0 \\ \cos x, & 0 \le x < \pi/2 \end{cases}$

12. $f(x) = \begin{cases} 0, & -2 < x < 0 \\ x, & 0 \le x < 1 \\ 1, & 1 \le x < 2 \end{cases}$

16. $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ e^x - 1, & 0 \le x < \pi \end{cases}$

Section 12.3 · ed. 5 p.669, p.670

In Problems 1–10, determine whether the given function is even, odd, or neither.

4. $f(x) = x^3 - 4x$ **6.** $f(x) = e^x - e^{-x}$

In Problems 11–24, expand the given function in an appropriate cosine or sine series.

12. $f(x) = \begin{cases} 1, & -2 < x < -1 \\ 0, & -1 < x < 1 \\ 1. & 1 < x < 2 \end{cases}$

In Problems 25–34, find the half-range cosine and sine expansions of the given function.

30. $f(x) = \begin{cases} 0, & 0 < x < \pi \\ x - \pi, & \pi \le x < 2\pi \end{cases}$ **33.** $f(x) = x^2 + x,\ 0 < x < 1$

Section 13.1 · ed. 5 p.692

In Problems 1–16, use separation of variables to find, if possible, product solutions for the given partial differential equation.

3. $u_x + u_y = u$ **8.** $y\dfrac{\partial^2 u}{\partial x \partial y} + u = 0$

In Problems 17–26, classify the given partial differential equation as hyperbolic, parabolic, or elliptic.

24. $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = u$

25. $a^2 \dfrac{\partial^2 u}{\partial x^2} = \dfrac{\partial^2 u}{\partial t^2}$

26. $k \dfrac{\partial^2 u}{\partial x^2} = \dfrac{\partial u}{\partial t}, \quad k > 0$

Section 13.2 · ed. 5 p.697

In Problems 1–6, a rod of length L coincides with the interval $[0, L]$ on the x-axis. Set up the boundary-value problem for the temperature $u(x, t)$.

3. The left end is held at temperature 100°, and there is heat transfer from the right end into the surrounding medium at temperature zero. The initial temperature is $f(x)$ throughout.

In Problems 7–10, a string of length L coincides with the interval $[0, L]$ on the x-axis. Set up the boundary-value problem for the displacement $u(x, t)$.

10. The ends are secured to the x-axis, and the string is initially at rest on that axis. An external vertical force proportional to the horizontal distance from the left end acts on the string for $t > 0$.

In Problems 11 and 12, set up the boundary-value problem for the steady-state temperature $u(x, y)$.

12. A semi-infinite plate coincides with the region defined by $0 \le x \le \pi$, $y \ge 0$. The left end is held at temperature e^{-y}, and the right end is held at temperature 100° for $0 < y \le 1$ and temperature zero for $y > 1$. The bottom of the plate is held at temperature $f(x)$.

Section 13.3 · ed. 5 p.699

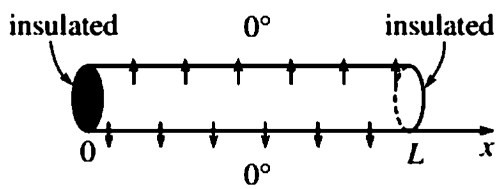
In Problems 1 and 2, solve the heat equation (1) subject to the given conditions. Assume a rod of length L.

2. $u(0, t) = 0, \quad u(L, t) = 0$
$u(x, 0) = x(L - x)$

5. Suppose heat is lost from the lateral surface of a thin rod of length L into a surrounding medium at temperature zero. If the linear law of heat transfer applies, then the heat equation

$$k\frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}, \quad 0 < x < L, \; t > 0,$$

h a constant. Find the temperature $u(x, t)$ if the initial temperature is $f(x)$ throughout and the ends $x = 0$ and $x = L$ are insulated.



heat transfer from lateral surface of the rod

Section 13.4 · ed. 5 p.703, p.704

In Problems 1–6, solve the wave equation (1) subject to the given conditions.

3. $u(0, t) = 0, \quad u(\pi, t) = 0, \quad t > 0$

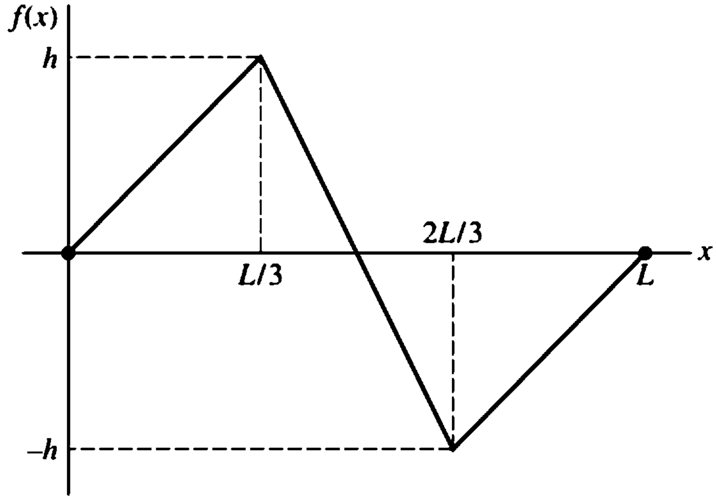
$$u(x, 0) = 0, \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = \sin x, \quad 0 < x < \pi$$

5. $u(0, t) = 0, \quad u(1, t) = 0, \quad t > 0$

$$u(x, 0) = x(1 - x), \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = x(1 - x), \quad 0 < x < 1$$

In Problems 7–10, a string is tied to the x-axis at $x = 0$ and at $x = L$ and its initial displacement $u(x, 0) = f(x)$, $0 < x < L$, is shown in the figure. Find $u(x, t)$ if the string is released from rest.

10.



Section 13.5 · ed. 5 p.709, p.710

In Problems 1–10, solve Laplace's equation (1) for a rectangular plate subject to the given boundary conditions.

4. $\left.\dfrac{\partial u}{\partial x}\right|_{x=0} = 0, \quad \left.\dfrac{\partial u}{\partial x}\right|_{x=a} = 0$

$u(x, 0) = x, \quad u(x, b) = 0$

10. $u(0, y) = 10y, \quad \left.\dfrac{\partial u}{\partial x}\right|_{x=1} = -1$

$u(x, 0) = 0, \quad u(x, 1) = 0$

In Problems 13 and 14, solve Laplace's equation (1) for a rectangular plate subject to the given boundary conditions.

13. $u(0, y) = 0, \quad u(a, y) = 0$

$u(x, 0) = f(x), \quad u(x, b) = g(x)$

Section 13.6 · ed. 5 p.716

In Problems 1 and 2, solve the heat equation $ku_{xx} = u_t, 0 < x < 1$, $t > 0$ subject to the given conditions.

1. $u(0, t) = 100, \quad u(1, t) = 100$
$u(x, 0) = 0$

6. Solve the boundary-value problem

$$k\frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}, \quad 0 < x < \pi, \ t > 0$$

$$u(0, t) = 0, \quad u(\pi, t) = u_0, \ t > 0$$

$$u(x, 0) = 0, \ 0 < x < \pi.$$

The PDE is a form of the heat equation when heat is lost by radiation from the lateral surface of a thin rod into a medium at temperature zero.

Section 15.3 · ed. 5 p.758

In Problems 1–6, find the Fourier integral representation of the given function.

2. $f(x) = \begin{cases} 0, & x < \pi \\ 4, & \pi < x < 2\pi \\ 0, & x > 2\pi \end{cases}$

In Problems 7–12, represent the given function by an appropriate cosine or sine integral.

10. $f(x) = \begin{cases} x, & |x| < \pi \\ 0, & |x| > \pi \end{cases}$